Continuous Valence-Corrected Intelligence

A Source-Grounded Representation of Intelligence as Situated Correction

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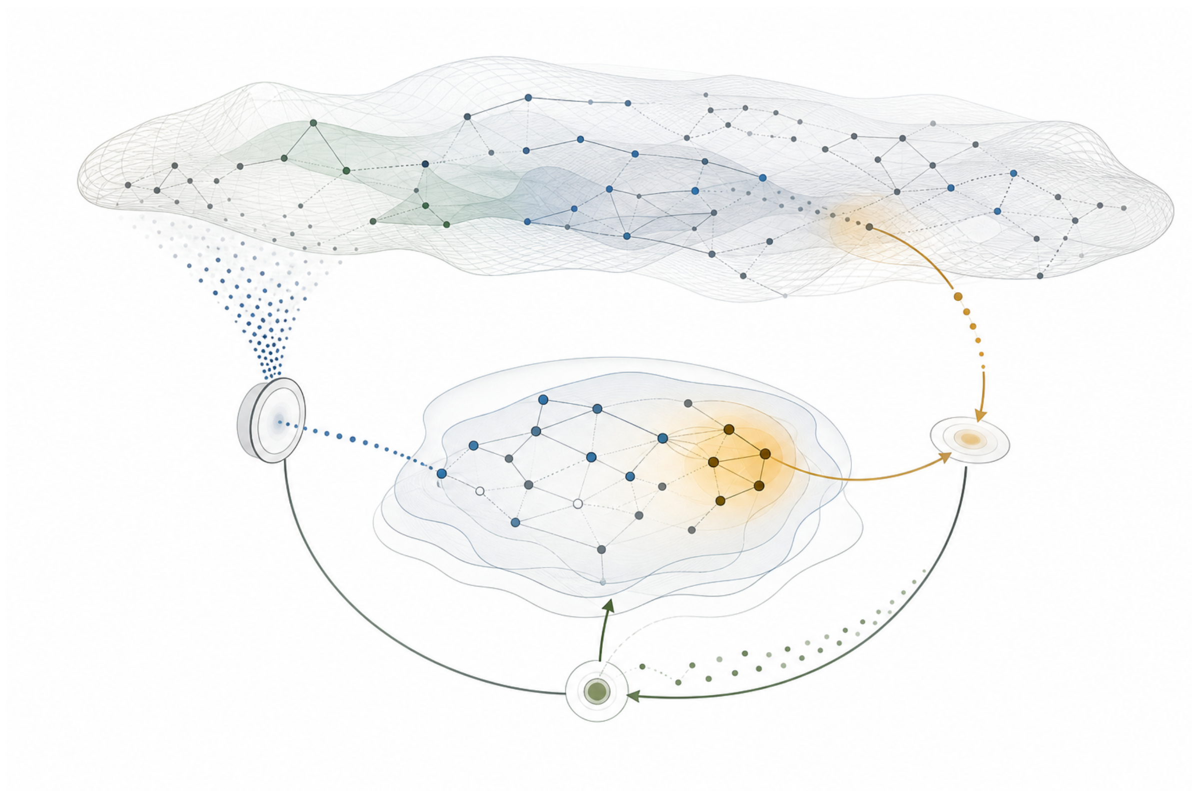


Conceptual supporting visual. Intelligence is pictured as situated correction: world structure is sampled, organized into internal representations, weighted by salience or value, tested through action, and revised through feedback. The correction loop is formalized in Fig. 1.

Abstract

This paper proposes *continuous valence-corrected intelligence*: a representation of intelligence as situated correction rather than reward maximization alone. The central hypothesis is that a system is generally intelligent to the degree that it can continuously revise a world model, allocate attention according to value or valence under uncertainty, and act in ways that expose and correct errors in both its model and its priorities. The proposal is motivated by four source families: learned world models for planning and imagination, continual learning under non-stationary streams, intrinsic-motivation signals such as prediction error and learning progress, and causal representation for transfer under intervention and distribution shift. Reinforcement learning is not rejected; the strongest reward-centered view already allows world models, prediction, planning, intrinsic rewards, and continual adaptation as mechanisms serving reward maximization. The paper therefore distinguishes reward maximization as a useful optimization method from reward maximization as an ontology of intelligence. It argues that reward is one correction signal inside a broader loop that also includes model error, salience selection, causal surprise, uncertainty reduction, value conflict, and stability constraints. The contribution is a source-grounded theoretical framework plus falsifiable evaluation protocols for testing whether value-shaped correction produces separable advantages over reward-only and intrinsic-reward baselines.

1 Introduction: The Wrong Representation of Intelligence

A representation of intelligence is not neutral. It decides which operations appear natural, which failures become visible, and which research directions seem inevitable. If intelligence is represented primarily as policy optimization over expected return, then the field is led toward better reward functions, stronger reinforcement learning pipelines, and more reliable optimization. These are useful tools, but the representation may be too narrow as an ontology.

The working hypothesis of this paper is that intelligence is better represented as a *situated correction process*. An intelligent system must operate inside a changing world, sample partial evidence, maintain an internal model of possibilities, decide what matters, act under uncertainty, and revise itself when the world pushes back. Reward may participate in that process, but reward is not the whole correction structure. The stronger proposal is that intelligence requires a loop in which world models, value, attention, and action correct one another over time.

The proposed representation is:

$$\boxed{I \;\sim\; \text{continuous world-model updating} + \text{value-shaped attention} + \text{self-correcting action}} \tag{1}$$

This formulation is intentionally a proposal, not an established theorem. Its purpose is to expose a research direction: intelligence should be studied as the maintenance and correction of an internal world model, modulated by value and tested through action. The phrase *valence-corrected* is used to emphasize that correction is not only epistemic. A system must correct what it predicts, but also what it attends to, what it treats as costly or worth preserving, and how it acts when those priorities meet the world.

The paper makes three contributions. First, it reframes intelligence as continuous correction rather than reward maximization alone. Second, it introduces value-shaped attention as a proposed bridge between world-model updating, intrinsic motivation, and action selection. Third, it proposes falsifiable evaluation settings for distinguishing continuous-correction agents from reward-only and intrinsic-reward baselines.

2 Background and Theoretical Pressures

The proposal is shaped by five source pressures. Sequence learning and world-model work show that agents can learn internal representations that compress temporal structure and support prediction, planning, or imagined behavior [1, 6, 7, 8]. Continual-learning work shows that agents learning from non-stationary streams face a stability/plasticity problem: they must adapt without overwriting useful prior knowledge [9, 10, 11]. Intrinsic-motivation work shows that internally generated signals such as prediction error, novelty, and learning progress can guide exploration when external reward is sparse, absent, rare, or deceptive [12, 13, 14]. Causal-representation work shows why robust generalization often requires distinguishing stable causal structure from spurious statistical association, especially when action changes future observations [2, 15, 16, 17]. Finally, the reward-centered view of Silver et al. gives the strongest pressure against the present theory by arguing that reward maximization may be sufficient to generate the abilities associated with intelligence in rich environments [19].

Table 1: How the cited source families constrain the proposed representation. The sources are used as claim boundaries, not as decorative citations.

Source family	**Source-backed claim**	**Constraint for this paper**
Sequence and world models	Learned internal representations can compress temporal or spatial structure and support prediction, planning, policy learning, or imagined rollouts [1, 6, 7, 8].	Intelligence should not be represented as raw input–output mapping alone. It requires updateable internal representation, but world models by themselves do not establish valence.
Continual learning	Learning from non-iid task streams creates forgetting, transfer, and stability/plasticity constraints [9, 10, 11].	Continuous updating needs retention and constrained plasticity; the theory cannot simply let model and value update freely.
Intrinsic motivation	Prediction error, novelty, and learning progress can guide exploration under sparse or absent external rewards [12, 13, 14].	Value-shaped attention should be framed as a hypothesis over correction-signal prioritization, not as established biological valence.
Causal representation	Robust transfer often requires separating causal structure from spurious association; agent actions can change future evidence [2, 15, 16, 17].	Action should be represented as intervention-like: it changes the data-generating process the agent subsequently observes.
Reward sufficiency pressure	A strong RL view can treat perception, memory, planning, language, intrinsic rewards, and world models as mechanisms serving reward maximization [19].	The paper must not caricature RL. It must show what continuous correction explains or predicts that reward-maximization ontology makes less visible.

2.1 World models and internal representation

Sutskever, Vinyals, and Le showed that a neural system can map variable-length sequences into internal representations and decode new sequences from them [1]. The result matters here less as a machine-translation result than as a representational event: a system can compress temporal structure into a learned state that supports later generation.

World-model-centered reinforcement learning gives a stronger technical foundation. Ha and Schmidhuber showed that an agent can learn a compressed spatial/temporal representation of an environment and use features from that model to train a compact controller, including training policies inside a learned latent "dream" environment [6]. PlaNet learns latent dynamics from image observations and uses them for online planning, supporting the claim that model-based agents can plan from learned latent structure in visual control settings [7]. Dreamer further separates world-model learning, behavior learning, and environment interaction, using latent imagination to learn long-horizon behavior [8].

These sources support a narrow claim: learned world models can compress experience, predict future dynamics or rewards, support planning or policy learning, and improve data efficiency in visual-control environments. They do not prove that intelligence as such requires valence-corrected world-model updating. They create the technical base on which the present hypothesis can be tested.

2.2 Continual learning and the stability/plasticity burden

Autonomous agents do not merely solve one stationary dataset. They encounter streams of changing evidence. Parisi et al. describe continual lifelong learning as a central problem for neural systems that must learn from non-stationary data while retaining prior knowledge [9]. Elastic Weight Consolidation gives one influential mechanism for slowing changes to parameters important for previous tasks, making the stability/plasticity dilemma concrete [10]. Gradient Episodic Memory formalizes continual-learning evaluation through task streams, transfer, and forgetting metrics [11].

For the present theory, the implication is direct: continuous correction is not automatically good. A system that updates too little becomes brittle; a system that updates too freely destroys useful structure. Any valence-corrected framework must specify which parts of the model, value system, and policy remain plastic, which are stabilized, and how correction signals are gated over time.

2.3 Intrinsic motivation and correction beyond external reward

Curiosity-driven exploration shows why external reward is not the only useful feedback signal. Pathak et al. operationalize curiosity as prediction error in a learned feature space, producing intrinsic reward when extrinsic reward is sparse or absent [12]. Random Network Distillation uses prediction error against a fixed random target network as a novelty signal for hard-exploration problems [13]. Oudeyer reviews computational theories of curiosity and emphasizes learning progress as a richer signal than raw novelty or surprise: agents should prefer regions where competence can improve, not regions that are already known or inherently chaotic [14].

These sources justify a careful claim: internally generated signals can guide exploration and representation learning. They do not justify saying that valence is already established as necessary for artificial intelligence. The safer formulation is that this paper proposes *value-shaped attention* as a unifying hypothesis over intrinsic correction mechanisms: prediction error, novelty, uncertainty reduction, learning progress, action relevance, and value conflict.

2.4 Causal agency and action as intervention

Causal inference emphasizes that statistical association is not the same as causal structure [2, 3]. Causal representation learning extends this pressure to machine learning by connecting causal variables and mechanisms to transfer, robustness, out-of-distribution generalization, and intervention [15]. De Haan et al. show a concrete agent-learning failure mode: imitation learners can rely on non-causal correlates in demonstrations and then fail under deployment shift, a phenomenon they call causal confusion [16]. Invariant Risk Minimization similarly pressures models to learn predictors that remain stable across environments rather than exploiting spurious correlations [17].

The present paper does not claim that every intelligent agent must explicitly contain a hand-authored structural causal model. The defensible claim is representational: action is intervention-like. It changes the future data-generating process the agent observes, making causal structure relevant to robust correction and transfer.

3 Relationship to Reinforcement Learning

Reinforcement learning gives a powerful mathematical frame: agents select actions to maximize expected return. This frame has produced indispensable algorithms, including model-based agents whose learned dynamics and value functions already resemble parts of the loop proposed here [7, 8]. The present paper is not anti-RL.

The strongest reward-centered counter-position is Silver, Singh, Precup, and Sutton's reward-is-enough hypothesis: intelligence and its associated abilities may be understood as subserving reward maximization by an agent acting in its environment [19]. On that view, rich environments can demand perception, knowledge, learning, social intelligence, language, generalization, imitation, memory, imagination, attention, reasoning, creativity, and emotions in the service of maximizing reward. The view also allows auxiliary mechanisms such as value prediction, model learning, planning, prediction of observations, and intrinsic rewards, while treating them as mechanisms serving future reward maximization.

This pressure matters. The paper cannot argue that reinforcement learning only optimizes external scalar rewards while the proposed theory handles models, attention, correction, and action. A serious reward-centered theorist can respond that those mechanisms can all emerge inside, or be engineered into, an agent whose ultimate objective is reward maximization.

The distinction this paper needs is therefore not *reward versus no reward*. The distinction is *method versus ontology*. Reward maximization can be a useful optimization method. Reward maximization can also be proposed as the ontology of intelligence. Continuous valence-corrected intelligence proposes a different ontology: intelligence is the maintenance of a correction loop in which reward is one signal among others.

Table 2: Two representations of intelligence. The proposed framework does not discard reward optimization; it relocates reward inside a broader correction loop.

Question	**Reward-maximization ontology**	**Continuous-correction ontology**
What is the agent?	A policy-bearing optimizer selecting actions to maximize expected return.	A situated system maintaining and revising a world model, value structure, attentional priorities, and action policy.
What is feedback?	Primarily reward or return, possibly supplemented by auxiliary objectives.	Any signal that corrects belief, salience, value, or action: reward, prediction error, causal surprise, consequence, conflict, uncertainty reduction, or stability pressure.
What is action?	An output chosen to improve expected return.	An intervention that tests the model, changes the world, and produces evidence for future correction.
What is value?	Usually encoded as reward, utility, return, preference, or value function.	A valence-like structure shaping what is attended to, preserved, avoided, explored, updated, or revised.
What is failure?	Low return, poor policy performance, or reward misspecification.	Breakdown in the correction loop: stale model, misplaced attention, unstable value update, poor intervention, reward overfitting, or destructive plasticity.

4 Proposed Framework

Let the world be denoted by $\mathcal{W}$, observations by o_t, actions by a_t, an internal world model by $\mathcal{M}_t$, a value or valence structure by $\mathcal{V}_t$, and an attention allocation by $\mathcal{A}_t$.

The proposed correction loop can be written schematically as:

$$\mathcal{A}_t = \text{Attend}(\mathcal{M}_t, \mathcal{V}_t, o_t), \tag{2}$$

$$a_t = \text{Act}(\mathcal{M}_t, \mathcal{A}_t, \mathcal{V}_t), \tag{3}$$

$$\mathcal{C}_t = \text{CorrectionSignal}(o_{t+1}, \mathcal{M}_t, a_t, \mathcal{V}_t), \tag{4}$$

$$(\mathcal{M}_{t+1}, \mathcal{V}_{t+1}) = \text{Update}(\mathcal{M}_t, \mathcal{V}_t, \mathcal{C}_t). \tag{5}$$

Here $\mathcal{C}_t$ is not restricted to reward. It may include prediction error, novelty, learning progress, causal surprise, failed expectation, constraint violation, value conflict, uncertainty reduction, observed consequence, or stability pressure. The central architectural claim is that the agent's next state should depend not only on what happened, but on how the event changes what is believed, what is salient, what is stable, and what is worth doing next.

This is not yet a full algorithm. It is a representation for algorithm design. It says that intelligence requires the co-evolution of model, value, attention, and action.

4.1 Continuous world-model updating

A world model is not a static database of facts. It is a continuously revised structure that makes future possibilities intelligible. World-model RL supports the plausibility of learned latent dynamics for planning and imagined behavior [6, 7, 8]. Continual learning adds the constraint that updating must be managed across non-stationary streams without catastrophic forgetting [9, 10, 11]. A frozen model may perform well inside its training distribution while failing when the world shifts; an unconstrained model may overwrite what made previous behavior competent.

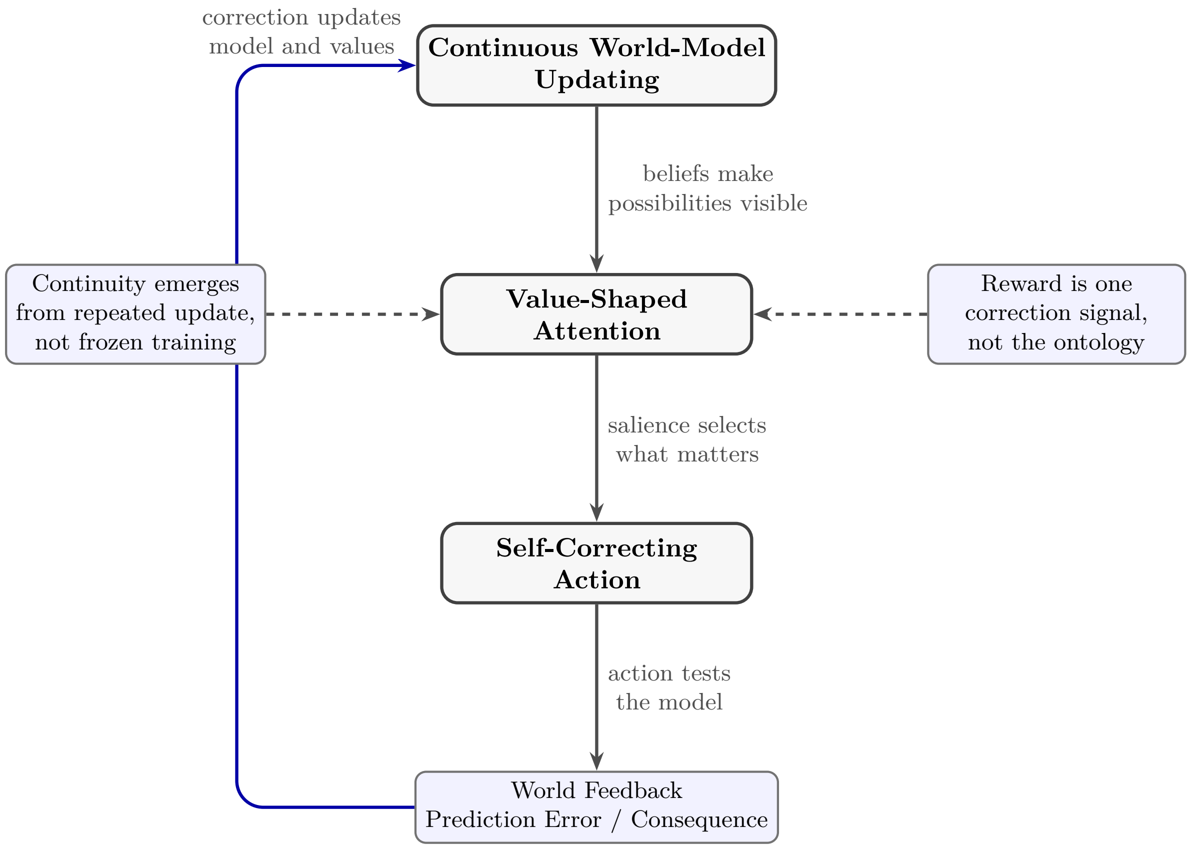
4.2 Value-shaped attention

Attention is not merely a computational convenience. In this proposal, value shapes attention by selecting which uncertainties, possibilities, or consequences matter. A system without value-shaped attention may perceive too much and understand too little. It may update indiscriminately, explore irrelevantly, or miss the features whose consequences matter most.

The word *value* is used broadly. It can include reward estimates, intrinsic costs, uncertainty reduction, preference, affect-like valence, safety constraints, and task commitments. The claim is not that human emotion must be copied literally. The claim is that valence-like structure may be central to competent agency because the agent must prioritize which errors, novelties, uncertainties, or conflicts deserve update priority.

4.3 Self-correcting action

Action is not only execution. Action is a test of the model. When an agent acts, the world returns consequences that reveal whether the model, values, and attentional choices were adequate. This gives action an epistemic role: it is an intervention-like operation that produces evidence. This claim is weaker than saying every agent requires an explicit causal graph, but stronger than treating action as mere output.



Proposed representation: intelligence is a living correction loop, not merely policy optimization over external reward.

Figure 1: The proposed formal representation of continuous valence-corrected intelligence. A world model makes possibilities visible, value-shaped attention selects what matters, self-correcting action tests the model, and world feedback updates both beliefs and values. Reward is treated as one correction signal rather than the ontology of intelligence.

5 Correction Signals Are Not Equivalent

A central risk for the framework is to rename existing reinforcement-learning machinery without adding explanatory power. To avoid that, the theory must separate correction signals by the operation they perform.

Table 3: Correction signals should be treated as operationally distinct. The table gives manuscript-safe distinctions; it does not claim that the boundaries are always clean in implemented agents.

Signal	**What it corrects**	**Primary risk**
Reward / return	Task success relative to an objective or preference proxy.	Reward misspecification, sparse feedback, or overfitting to the proxy.
Prediction error	Mismatch between expected and observed consequence.	Chasing stochastic or unlearnable noise.
Novelty	Lack of prior familiarity with states or features.	Exploring irrelevant novelty rather than useful structure.
Learning progress	Improvement in prediction, control, or competence.	Ignoring rare but important states where progress is initially slow.
Causal surprise	Failure of an intervention-relevant model under action or distribution shift.	Requiring causal variables that are not yet identifiable.
Value conflict	Tension between competing goals, constraints, or priorities.	Destabilizing values if conflict resolution is unconstrained.
Stability pressure	Protection of previously useful knowledge or commitments.	Excessive rigidity and failure to adapt.

This separation creates the paper's testable burden. If all of these signals can be reduced without loss to reward maximization in rich environments, then the proposed representation may be merely a descriptive relabeling. If, however, freezing or perturbing model, value, attention, and correction channels produces separable failures, then continuous correction becomes more than rhetoric.

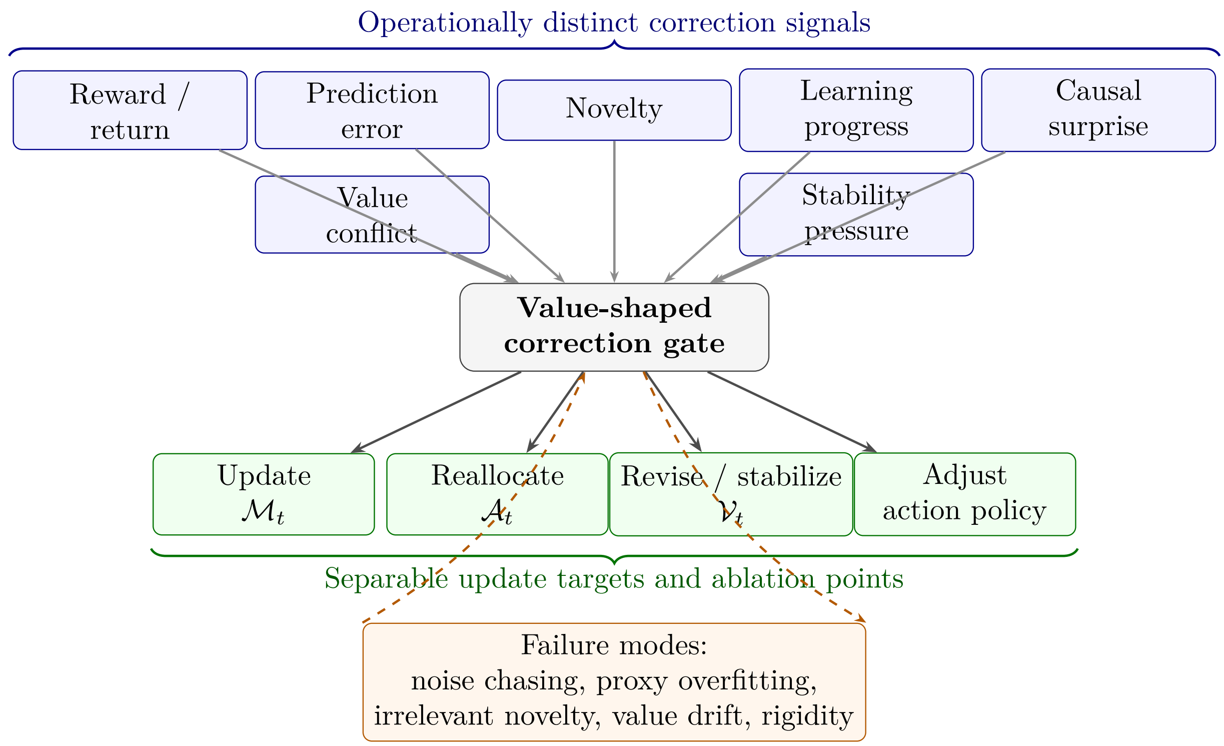


Figure 2: Correction signals are not treated as interchangeable reward aliases. Distinct signals enter a value-shaped correction gate that can update different targets: the world model, attention allocation, value structure, or action policy. The lower feedback loop marks the main risks: proxy overfitting, noise chasing, irrelevant novelty, value drift, and excessive rigidity.

6 Training Implications: Developmental World-Model Agents

If the representation is right, then the next training paradigm should be developmental rather than merely episodic. A developmental agent would be trained not only to maximize task reward, but to improve the loop itself: model updating, salience allocation, intervention selection, and correction under distribution shift.

A practical research program would include:

1. **Continual model revision:** agents must update internal representations without catastrophic collapse.
2. **Value-attentional learning:** agents must learn what to inspect before external reward is available.
3. **Intervention-aware action:** actions should improve future model quality, not only immediate return.
4. **Multi-timescale correction:** fast updates should handle local surprise while slower updates reshape values, commitments, and abstractions.
5. **Causal generalization:** agents should transfer across contexts by preserving intervention-relevant structure.

URLB suggests a useful two-phase evaluation structure: reward-free pretraining with intrinsic rewards followed by downstream adaptation under extrinsic rewards [18]. The present framework can adopt this structure while adding ablations for value-shaped attention, causal transfer, and stability/plasticity.

7 Evaluation: Falsifiable Predictions and Protocols

The proposal should be judged by tests that can fail. Each prediction below names a setup, measurement, baseline, and failure condition.

Table 4: Falsifiable prediction matrix for continuous valence-corrected intelligence. Each prediction is designed to distinguish the proposed representation from reward-only or intrinsic-reward baselines.

Prediction	**Experimental setup**	**Measurement**	**Baselines**	**Failure condition**
Salience-shift adaptation	Partially observable environment where reward remains fixed but causally relevant features shift across episodes.	Regret, recovery time, and trials to adaptation after each salience shift.	Reward-only RL; novelty-only intrinsic reward; world-model RL without value-shaped attention.	Baselines match adaptation without the proposed mechanism.
Separable freeze failures	Freeze $\mathcal{M}_t$, $\mathcal{V}_t$, or $\mathcal{A}_t$ separately while the rest of the agent adapts.	Error profiles: prediction/planning failure, exploration misallocation, or value-conflict failure.	Full adaptive agent; single-channel ablations.	Freezing channels produces indistinguishable failures.
Causal transfer	Train with spurious correlations that break during transfer while causal mechanisms remain stable.	OOD return, intervention success rate, and reliance on spurious features.	Empirical risk or reward-optimized agents without intervention-aware training.	Intervention-aware correction shows no transfer advantage.
Dense correction efficiency	Compare terminal reward alone against terminal reward plus structured correction signals.	Sample efficiency, robustness under delayed feedback, downstream adaptation return.	Extrinsic reward only; prediction error only; novelty only; learning progress only.	Structured correction does not improve robustness or efficiency.
Pretraining/adaptation robustness	Use URLB-style reward-free pretraining followed by fixed-budget downstream adaptation.	Normalized downstream return, adaptation slope, monotonicity with pretraining budget, pixel/state gap.	Existing unsupervised RL baselines and world-model baselines.	Gains vanish, degrade with pretraining, or remain task-specific.

7.1 Protocol 1: reward-free pretraining followed by fixed-budget adaptation

The cleanest first protocol borrows URLB's structure [18]. During pretraining, the agent interacts without task-specific extrinsic reward. The training signal is intrinsic or correction-based: prediction error, novelty, learning progress, uncertainty reduction, or the proposed value-shaped correction. During downstream adaptation, the agent receives extrinsic reward for a fixed budget. The key measurements are normalized downstream return, adaptation speed, performance as a function of pretraining steps, and failure cases where more pretraining harms adaptation.

7.2 Protocol 2: salience-shift environments

In salience-shift environments, the reward function remains fixed while the observation features relevant for success change across episodes. The theory predicts that agents with value-shaped attention should recover faster than agents that rely on reward-only optimization or undifferentiated novelty. This protocol tests whether the proposed mechanism helps decide which uncertainty matters, not merely whether the agent can explore.

7.3 Protocol 3: separable model/value/attention ablations

The strongest internal test is ablation. If freezing the world model, value structure, or attention allocator produces distinct error profiles, then the framework has separable mechanism content. If all freezes produce the same performance degradation, or if reward-centered baselines absorb the difference cheaply, the theory is weakened.

7.4 Protocol 4: causal transfer under broken correlations

Following the pressure from causal representation learning, causal confusion, and invariant-risk work [15, 16, 17], agents should be tested in environments where spurious correlations predict reward during training but break during deployment. The predicted advantage is not generic OOD robustness; it is improved transfer when action-relevant causal structure remains stable while observational shortcuts fail.

8 Alignment Implications

The framework suggests that alignment is not only a reward-specification problem. It is also a question of what the system attends to, how it updates values, and how feedback changes its internal model over time. A system that optimizes a fixed objective without revising its understanding of consequences may be brittle or dangerous. A system that updates without stable value structure may drift.

The alignment-relevant target is therefore a stable but corrigible correction loop: values must shape attention and action, but feedback must be able to expose when those values or their operational proxies are wrong. Continual-learning work makes the stability problem technical rather than rhetorical [9, 10]. Intrinsic-motivation work adds a second safety pressure: exploration signals can produce behavior that is useful for learning but misaligned with human intent unless constrained [13, 18].

9 Limitations and Open Questions

This paper proposes a representation and research agenda. It does not provide a finished algorithm, theorem, or empirical validation. Several limitations are load-bearing:

- **Valence is a hypothesis.** The cited intrinsic-motivation literature supports prediction error, novelty, and learning progress as useful signals. It does not prove biological valence or affect is computationally necessary.

- **Reward may subsume the proposed mechanisms.** A reward-centered agent can include world models, planning, intrinsic rewards, attention, and continual adaptation [19]. The theory must earn its distinction through separable predictions.
- **World models need salience.** Learned models can encode irrelevant details or miss task-relevant structure. The paper proposes value-shaped attention as a possible solution, not an established one.
- **Continuous updating is dangerous.** Stability/plasticity, catastrophic forgetting, and value drift remain unsolved constraints [9, 10, 11].
- **Causal variables are not given.** Causal representation learning remains an open problem; the framework cannot assume that intervention-ready variables are already available [15].
- **Benchmark transfer is uncertain.** Existing unsupervised RL benchmarks show that intrinsic pretraining is promising but insufficient; more pretraining does not reliably improve downstream adaptation [18].

These limitations are not cosmetic. They define the experimental path by which the theory must either mature or fail.

10 Conclusion

The central claim of this paper is that intelligence is better represented as continuous situated correction than as reward maximization alone. A generally intelligent system maintains a world model, allocates attention according to value, acts to test and change the world, and updates itself from the consequences. Reinforcement learning remains useful, but it becomes one part of a broader loop.

The paper's contribution is representational. It proposes that reward, prediction error, novelty, learning progress, causal surprise, value conflict, and stability pressure should not be collapsed prematurely. They are correction signals with different roles in an agent's ongoing adaptation. If that distinction yields separable empirical predictions, continuous valence-corrected intelligence becomes a viable research program. If reward-centered agents absorb the distinction without loss, the theory should be revised.

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